Hydraulic Press Preventive Maintenance Checklist

Overview

The hydraulic press at Riverbend Packaging is a 50-ton downstroking press used to form and bale packaging scrap and to press fit components during repairs. This checklist defines the preventive maintenance tasks, intervals, and acceptance criteria that keep the press safe and reliable.

Specifications

Item	Value
Capacity	50 tons
System pressure	2,000 psi maximum
Reservoir	40 gallons, ISO VG 46 hydraulic oil
Motor	15 hp, 460 V
Stroke	24 inches
Filter	10 micron return line filter

Before Any Maintenance

> **WARNING:** The press stores energy in the hydraulic system and in the raised ram. Lock out MD-8 and hydraulic valve HV-1, bleed accumulator pressure to zero, and install the mechanical ram block before placing any part of the body under the ram.

Follow the Lockout/Tagout Procedure and wear leather gloves, safety glasses, and hearing protection.

Daily Checklist

1. Check the oil level in the sight glass; it should be between the marks.
2. Inspect hoses and fittings for leaks or weeping.
3. Check the return filter indicator.
4. Test the two-hand controls and emergency stop.
5. Look for oil on the floor and clean it up immediately.
6. Confirm guards and light curtains are in place and functional.

Weekly Checklist

1. Check system pressure against the gauge and the setpoint of 1,800 psi.
2. Lubricate guide posts with way oil.
3. Inspect electrical enclosures for moisture or loose wires.
4. Verify that the ram block stored on the press is intact.

5 Record cycle count.

Monthly and Longer Intervals

Interval	Task	Acceptance criteria
Monthly	Test pressure relief valve	Opens at 2,000 psi plus or minus 50
Monthly	Inspect cylinder rod for scoring	No visible scoring or pitting
Quarterly	Sample hydraulic oil for contamination	Cleanliness ISO 18/16/13 or better
Quarterly	Replace return filter element	Element changed regardless of indicator
Semi-annual	Inspect hoses for cracking	Replace if cracked or bulging
Annual	Replace oil if sample fails	Drain and refill 40 gallons
Annual	Inspect press frame welds	No cracks
Annual	Calibrate pressure gauge	Within 2 percent of reference

Troubleshooting

Symptom	Likely cause	Corrective action
Slow ram speed	Worn pump or low oil	Check oil; test pump flow
Press will not build pressure	Relief valve stuck open	Clean or replace relief valve
Oil overheating	Low oil or blocked cooler	Check level; clean cooler
Oil leak at cylinder	Worn rod seal	Replace seal kit
Ram drifts down	Leaking valve or seal	Test and replace valve
Excessive noise	Air in system or cavitation	Bleed system; check suction

Oil Changes and Disposal

1 Lock out the press and install the ram block.

2 Drain the reservoir into a covered container.

3 Clean the tank and replace the suction strainer.

4 Refill with new ISO VG 46 oil through a filter cart.

5 Bleed air by cycling the ram without load.

6 Dispose of used oil through the approved waste vendor.

Hydraulic oil spills create a slip hazard and must be cleaned up with oil-only absorbent.

Records

Technicians initial each task on the checklist and record anomalies. Out-of-tolerance findings are escalated to the Maintenance Supervisor before the press is returned to service. Maintain checklists for two years.